

WEEK 2

Frequency Throttling Side Channel Attack

SUPERVISORS

Prof Abhishek Bichhawatt
Subhrajit Das

Ramji Purwar
24110287

Experiments

Experiment 1

Effect of Workload and Active Cores on Frequency

Change the amount of work and the number of active cores, then observe effective CPU frequency.

Experiment 2

Data-Dependent CPU Power

Keep the computational workload essentially the same but change the input data, then observe power.

Hardware and common setup

- CPU: Intel Xeon Gold 5512U (Emerald Rapids).
- 28 physical cores / 56 threads; experiments use one thread per physical core.
- Workers are pinned to selected physical cores using CPU affinity.
- Synchronization barriers are used so workers begin together.
- Measurements are designed to observe physical CPU behavior rather than relying only on OS-level requests.

Effect of Workload and Active Cores on Frequency

How does CPU frequency respond when we change:

- the amount of computational work (number of instructions / burst size)?
- the number of active CPU cores?

WorkLoad

- AVX-512 FMA was chosen because it creates high computational and power demand, making frequency effects easier to expose.
- One 512-bit ZMM register contains 16 single-precision floats.
- Eight independent accumulator chains are used to keep the FMA execution resources highly utilized.
- Data is kept in registers during the measurement to avoid memory stalls.

What we changed

Variable 1 — Workload size: burst size increased from small workloads up to 50 billion instructions.

Variable 2 — Active cores: 1, 2, 4, 6, ..., up to 28 physical cores.

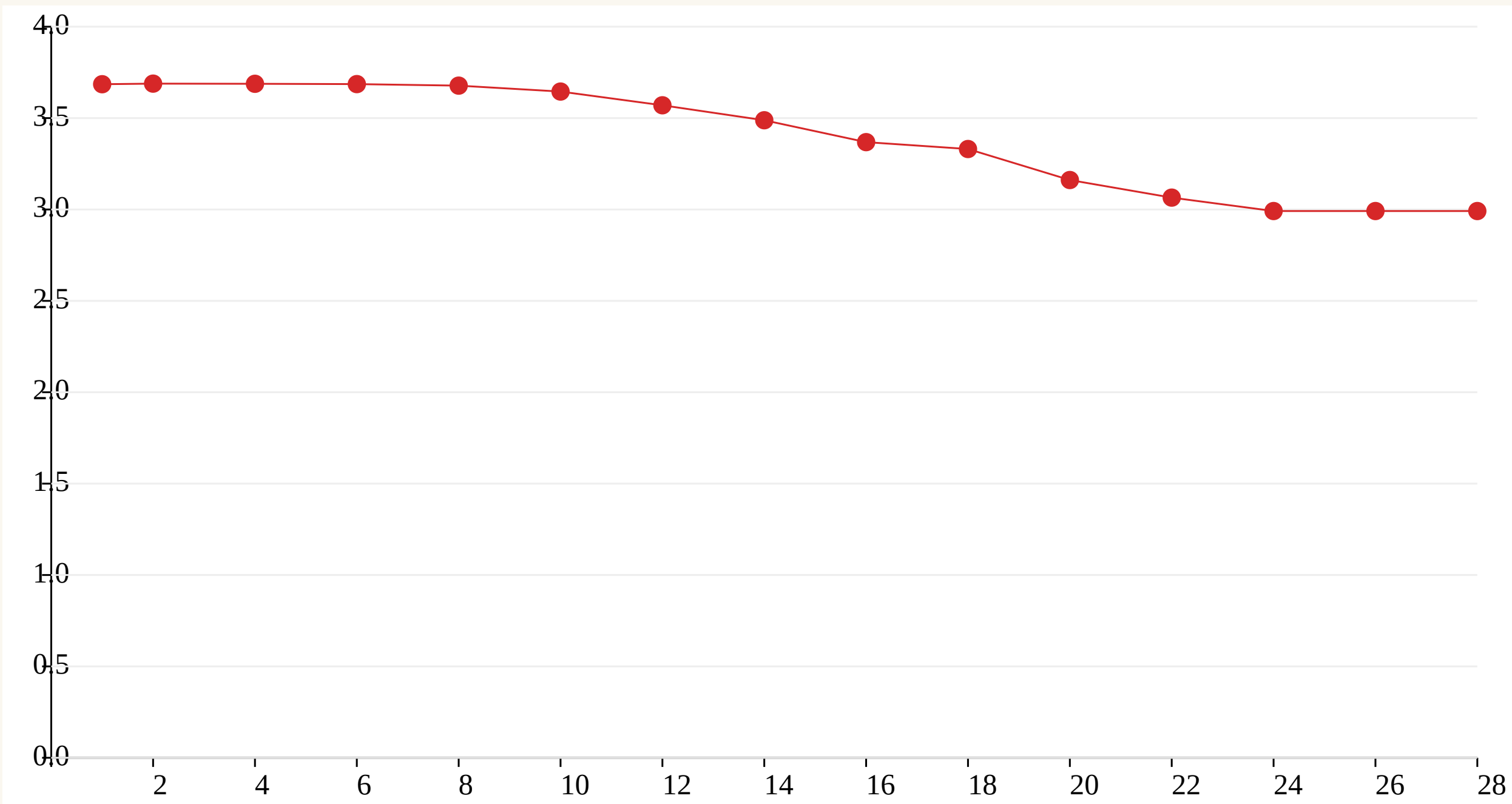
Result: Changing workload size

- At a fixed number of active cores, increasing the number of instructions increased execution time.
- However, effective frequency remained approximately constant across the tested burst sizes.
- Therefore, simply making the workload longer did not produce progressive frequency reduction in this experiment

Observation:

More instructions → more execution time → approximately the same frequency.

Result: Changing Active Cores



- When more physical cores execute the high-power workload simultaneously, total package demand increases.
- Frequency decreases as the number of active cores increases.
- At 28 active cores, the observed sustainable frequency is approximately 2.9–3.0 GHz.

Observation:

More active cores → higher aggregate power demand → lower sustainable frequency.

Data-Dependent CPU Power

Can changing only the input data change CPU power consumption?

- Keep the instruction sequence, number of operations, number of cores, and thread placement constant.
- Use two different 64-bit operands for the same repeated IMUL workload.
- Measure package and DRAM energy using Intel RAPL and derive average power from energy divided by execution time.

Workload and controls

- 28 physical cores; one worker thread per core.
- Approximately 2.8 billion IMUL instructions per core.
- Four independent 64-bit registers are repeatedly multiplied by the supplied operand.
- Same instruction sequence for both inputs: repeated imulq operations.
- Five warm-up trials and 300 measurement trials per operand.
- Threads are pinned and synchronized using barriers.

Data-Dependent CPU Power

What data being changed

Input A (0x00): 0x00000000000000000000

Input B (labeled 0xAA): 0xAAAAAAAAAAAAAAAAAAB

Only the operand data is changed between the two workloads.

Note: The current B value ends in B, so it is not literally 0xAAAAAAAAAAAAAAAAAA.

Measurement

- Execution time is measured using CLOCK_MONOTONIC_RAW.
- RAPL energy is sampled immediately before and after the synchronized workload.
- Package energy difference gives the energy consumed during the measurement interval.
- Average package power = package energy / mean execution time.
- Package - DRAM is treated as non-DRAM package energy; it is not a pure CPU-core power measurement.

Results

Data : 0x00

Avg Execution Time : **0.9358 s**

Avg Power Consumed : **133.8 W**

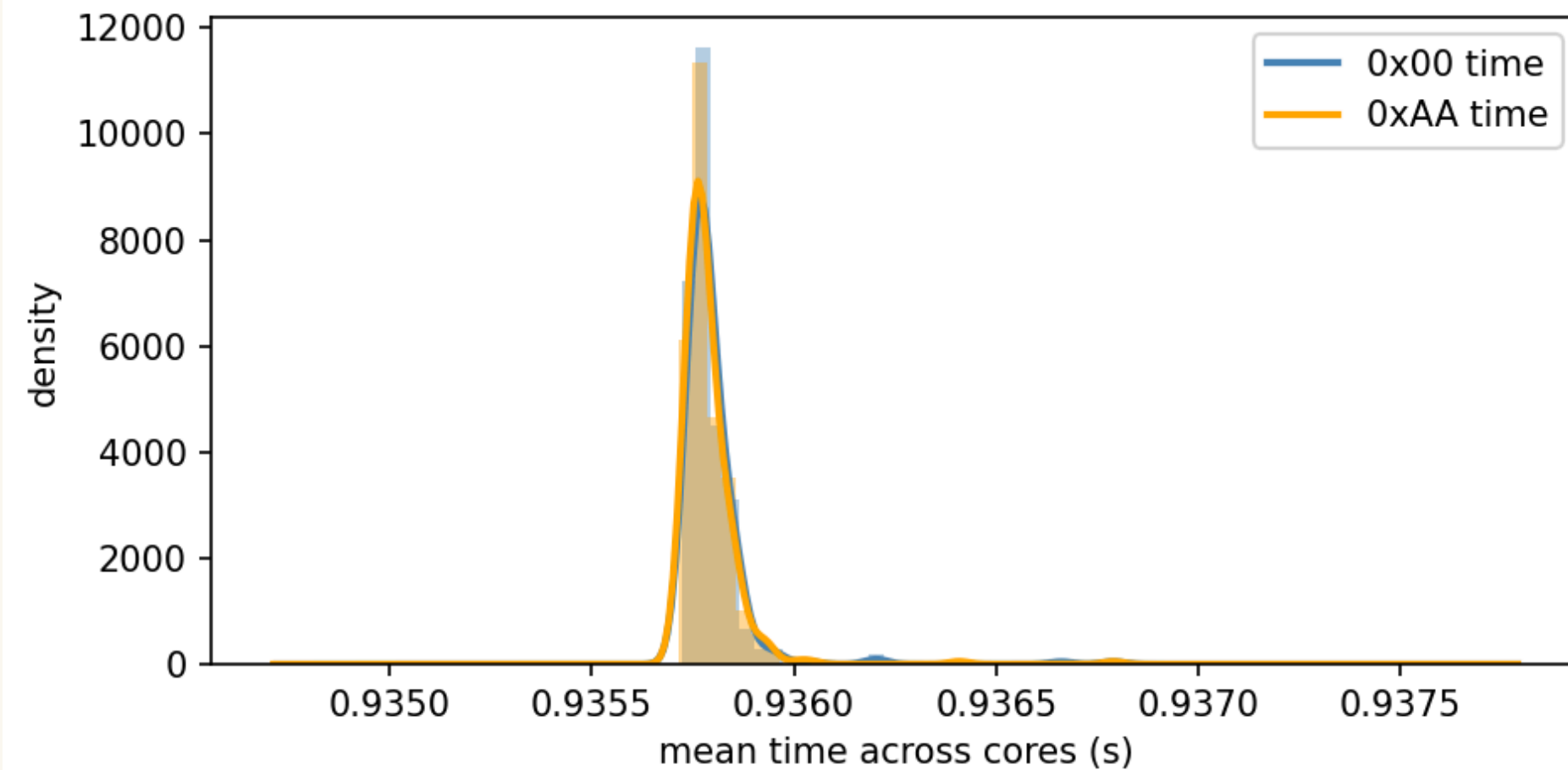
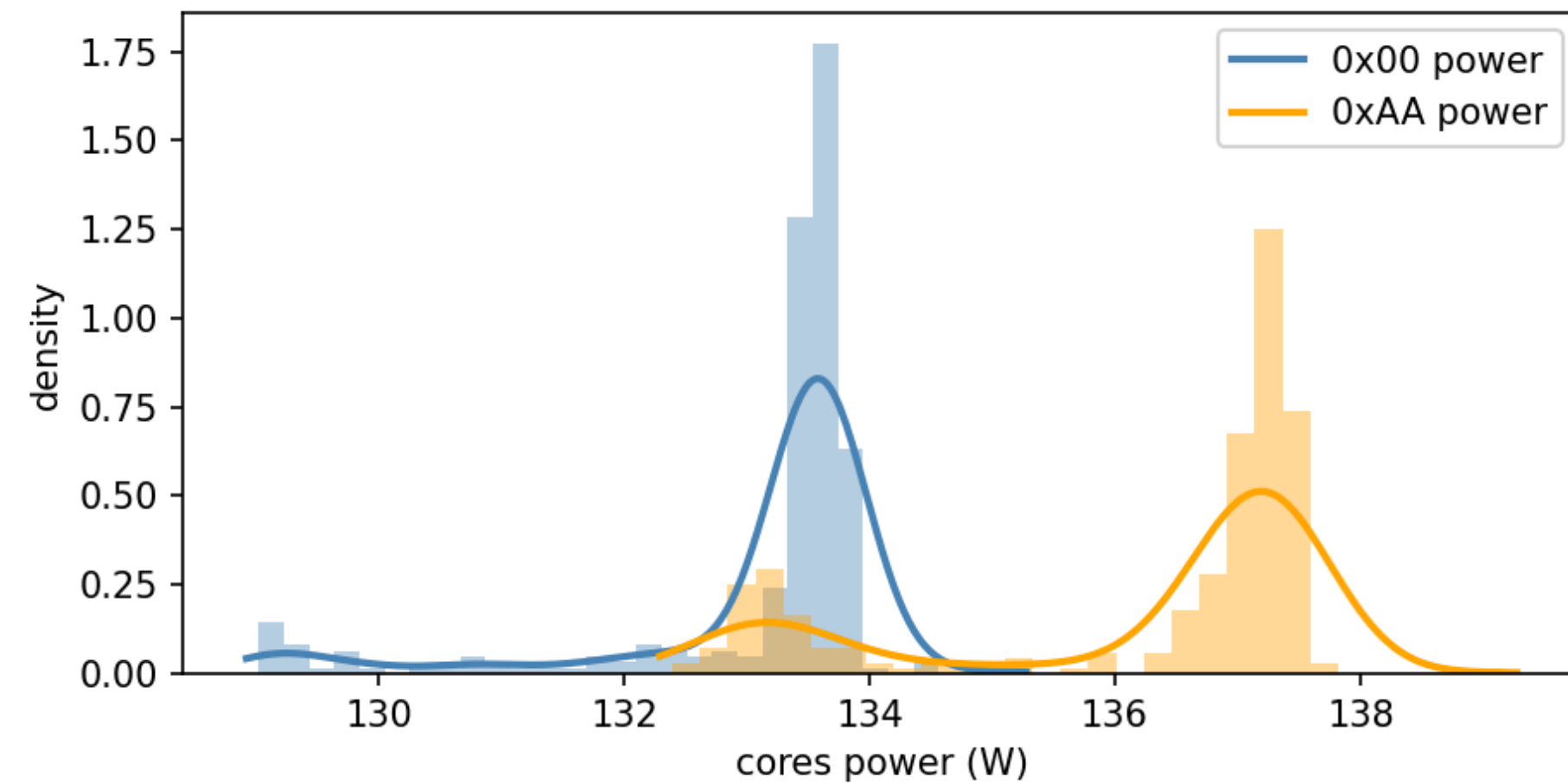
Data : 0xAA

Avg Execution Time : **0.9358 s**

Avg Power Consumed : **137 W**

Power distributions are clearly separated: the 0xAA workload consumes more measured power.

Execution-time distributions overlap heavily.
Thus the power difference cannot be explained simply by one workload doing substantially more work.



Physical interpretation

Dynamic CMOS power is approximately related to switching activity:

$$P_{\text{dynamic}} \propto \alpha C V^2 f$$

- Changing operand bits can change internal circuit transitions and switching activity α .
- Therefore, identical instructions can have different physical power costs depending on the data.
- However, RAPL measures package-level energy, so this experiment does not isolate the integer multiplier as the sole source of the difference.

The End